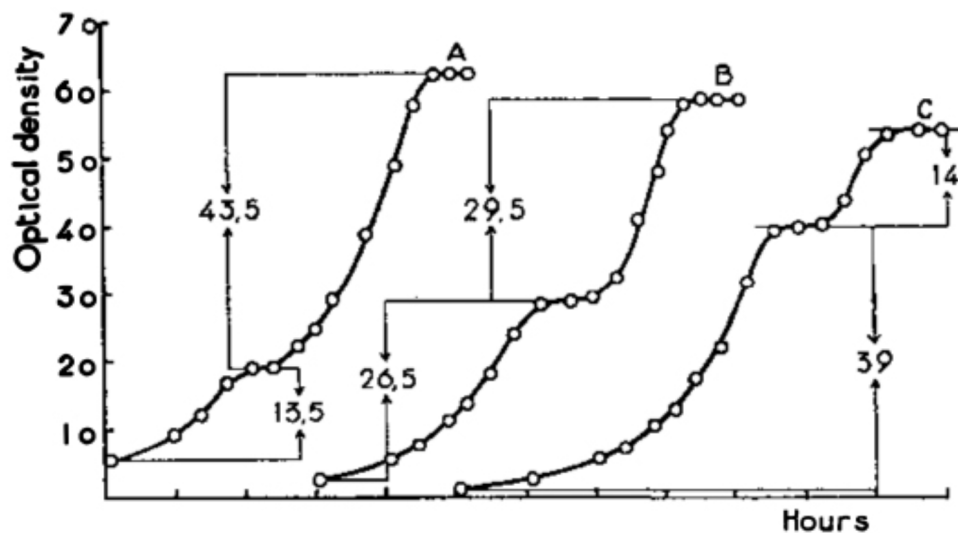


Lecture 14 Live Problems

The lac operon: a biological switch

Points toward lab manual pages: **Lac1, Lac2, Lac3, Lac4** (section L3 S3)

1. Diauxic growth. The figure below shows the growth of *E. coli* populations under three conditions: (A) 50 $\mu\text{g/mL}$ glucose, 150 $\mu\text{g/mL}$ sorbitol; (B) 100 $\mu\text{g/mL}$ glucose, 100 $\mu\text{g/mL}$ sorbitol; (C) 150 $\mu\text{g/mL}$ glucose, 50 $\mu\text{g/mL}$ sorbitol.

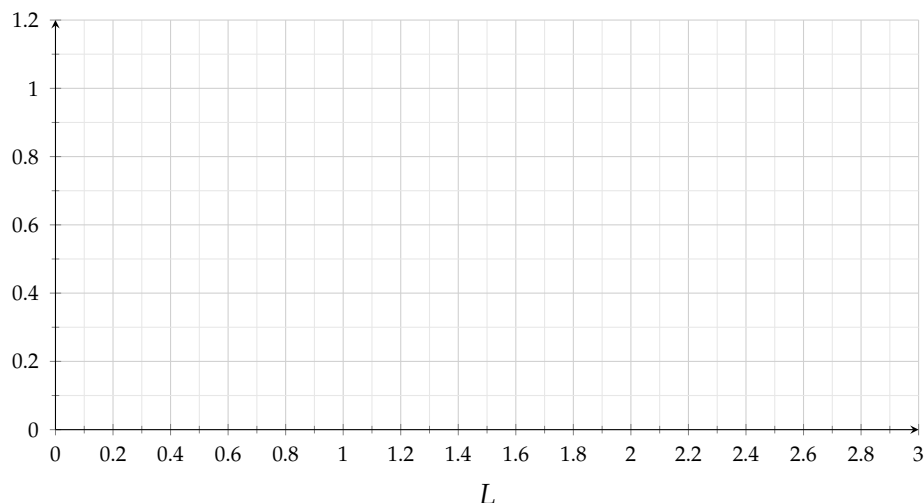


Monod, J. (1942), *Recherches sur la croissance des cultures bactériennes*.

What does this figure tell us?

2. Equilibria of the lactose switch. Find the equilibrium point(s) for the change equation

$$L' = 0.04 + \frac{L^2}{1 + L^2} - 0.4 L.$$



3. Adding glucose. How do the equilibrium points change when glucose g increases?

$$L' = 0.04 + \frac{L^2}{1 + (1 + g) L^2} - 0.4 L.$$

Glucose g	Phase portrait
0	
0.5	
1	
2	